SUNANDITHA B S

sunanditha24@gmail.com | Portfolio | LinkedIn | GitHub

Education

RNS Institute of Technology, Bengaluru CGPA: 8.9
B.E. Information Science & Engineering

Technical Skills

Languages: C, Python, Bash, SQL **Systems:** Linux, Git, Docker **Embedded and Automotive:** CAN, SocketCAN, UDS (ISO 14229) **Security:** X.509, RSA-PSS, Secure Boot, OTA

Systems Engineering Experience

- Diagnosed and resolved two Linux boot failures via live-USB chroot recovery: a GPU driver corrupting init, and an Intel VMD/RST controller loading out of order. Fixed both with initramfs/GRUB rebuilds and module reordering.
- Worked out GPT geometry by hand at the sector level to move immovable NTFS $MFT mirror metadata, rebuilding partitions for a 60% capacity increase with zero data loss.
- Reverse-engineered a MediaTek boot trust chain (BROM → Preloader → LK → AVB) to load a custom system image on a locked bootloader, patching vbmeta flags and using fastbootd.
- Tracked down a 4-layer networking failure (TLS, IPv6, DNS, socket permissions) blocking a containerized FastAPI deployment on Docker, isolating and fixing each layer end-to-end.

Projects

BOLT: Boot & OTA Lockdown Toolkit Jun – Jul 2026
Python | RSA-PSS/SHA-256 | X.509 | Uptane-inspired OTA | pytest

- Designed a secure OTA and boot integrity architecture for automotive ECUs: signed firmware images validated through a 2-tier X.509 trust chain, with A/B partitioning so a crash mid-flash never corrupts the active image.
- Wrote 4 attacker modules (rollback replay, signature stripping, MITM, power-loss truncation) to red-team the design; all 4 got blocked by the trust chain and rollback protection.
- Verified the framework with custom-built pytest suite for 33 tests covering signing, verification, partition switching, and every attack path.

CANary: CAN Bus Intrusion Detection Framework July 2026
Python | C | SocketCAN | ctypes | pytest

- Designed and implemented an 8-module CAN bus IDS around priority-based arbitration and a native 15-bit CRC checksum, with the C core bridged to Python via ctypes across 3+ simulated ECUs.
- Wrote 6 rule-based detectors for spoofing, replay, flooding, and malformed frames, surfaced through 4-tier severity alerts. 47 tests pass on live SocketCAN.

WARDEN: UDS Diagnostic Session Security Simulator May 2026
Python | SocketCAN | UDS (ISO 14229) | pytest

- Built a UDS (ISO 14229) session simulator over SocketCAN with seed-key SecurityAccess and attempt lockout, then wrote an attacker client that brute-forces the key.
- Detector tracks DiagnosticSessionControl transitions and SecurityAccess timing, flagging repeated key attempts and negative response codes inside a sliding window. 9 tests cover session logic and malformed input.

Research

BICE: Behavioral Identity Continuity Engine Feb 2026 – Present
Under Review – ICICKE 2026

- Built a per-device drift-based anomaly detector for IoT telemetry, using sliding-window Z-scores on Welford online statistics for constant-memory O(1) updates across N-BaIoT's 100-feature space.
- Evaluated on 89 profiles from 9 IoT devices and got 81.2% TPR at 0% FPR, ahead of both Isolation Forest and One-Class SVM baselines.

Achievements

TryHackMe Top 4% | Winner, Escape and Exploit Cybersecurity Competition | Top 15, ZeroDay Arena CTF